



The Break — Empty Set

Artist's Proof 09

Quantum Mechanics

Superposition, measurement, the Born rule, the Schrödinger equation

0 — Dependency and Scope

What this paper does

AP09 shows that quantum mechanics — superposition, measurement, entanglement, the Born rule, the Schrödinger equation — follows from the same four axioms {S, B, R, C} and bridge hypotheses (EH, QRA) that produce spacetime (AP10) and gravity (AP08).

The derivation proceeds through structural readings and mathematical derivations.

Structural readings identify the pre-state with superposition, the break with measurement, and unbroken degrees of freedom with entanglement.

Mathematical derivations produce linearity, the Born rule, and the Schrödinger equation via uniqueness theorems.

Epistemic status is declared per section. Some results are derived. Some are structural readings. Some are bridge identifications. The distinction is maintained throughout.

No part of the derivation assumes the postulates of quantum mechanics.

Dependencies

AP20 (The Proof): the axioms {S, B, R, C} — completeness (AP20 4.1, fenced KS-P.1), independence and minimality (AP20 4.2, fenced KS-P.2), joint consistency from the undeniable premise (AP20 6). Load-bearing.

AP10 4 (The Dimension): Under EH + QRA, Lorentzian manifold (M, g) with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$ — the signature read from the axioms' reversibility structure, formal metric-level derivation fenced at KS-D.4. Load-bearing for complex amplitudes (the complex amplitudes derivation) and the Schrödinger equation (the Schrödinger derivation).

AP06 (The Leakage Constant): $\varepsilon > 0$ whenever c is finite. Referenced for entanglement (in the entanglement section) and the eye topology.

AP08 (The Identity): Einstein's field equations from the record algebra. Referenced for the gravity–quantum duality and conservation laws.

AP20 (The Proof): EH and QRA proven. All results that depended on EH are now unconditional.

The Lock (Edition 04): ε = electron. Load-bearing for the identification of the break element with a physical particle.

Epistemic status per phase

Foundational phase (empty set ontology, the break). Structural reading of the pre-state and break — ontological identification of $\{S, B\}$ with the empty set splitting. Not a derivation; a reading that grounds the rest. KS-Q.10.

Linearity construction (the bridge step). Bridge derivation. Linearity constructed from the record monoid (R) + complex scalars (Lorentzian signature) + faithful embedding (EH). Compatibility of monoid action with vector space operations. KS-Q.7.

Complex amplitudes (the conditional derivation). Derivation conditional on reconstruction theorems. Complex amplitudes from the Lorentzian signature. KS-Q.4.

Measurement. Structural reading. Measurement as the break. Irreversibility from Axiom R. The loop from the axiom structure. KS-Q.5.

Entanglement. Structural reading. Entanglement as unbroken pre-state. The Bell correlation account is compatibility, not derivation. KS-Q.3.

Born rule (the conditional derivation). Derivation. Born rule from Axiom S. $\sigma \leftrightarrow$ conjugation proven unique (KS-Q.6 CLOSED). Conditionally closed (conditional on KS-Q.7). KS-Q.1, KS-Q.2.

Ergodicity (the structural extrapolation). Everything possible actualises across infinite loops. Structurally motivated but formally unproven. KS-Q.9.

Schrödinger equation (the strongest derivation). Derivation. Via Wigner + Stone applied to the derived Hilbert space. $\hbar$ identified, not derived in this paper (KS-Q.8 — closed conditional on KS-L.4 by AP12).

Axiom mapping

Axiom S (Symmetry). The pre-state. The 1:1. Two sectors indistinguishable before the break. S is the superposition (read in the empty set and break sections). 0 and 1 co-existing.

S provides the involution σ which, under the Born rule derivation (later in the paper), maps to complex conjugation.

Axiom B (Break — unique). The measurement. One element ε with no σ -image. The first record.

B is the act of distinguishing (read in the break section): superposition becomes definite outcome. Under The Lock, ε = electron.

Axiom R (Record — monotone). Irreversibility. The monoid gives the arrow of time.

R is what makes measurement irreversible (read in the measurement section). Once a record is written, it cannot be unwritten.

R provides the Hilbert space structure through the monoid's sequential composition.

Axiom C (Constraint). No-signalling. The finite rate c prevents instantaneous correlation exploitation (in the no-signalling argument within the entanglement section).

C provides the Lorentzian structure from which complex amplitudes are derived (in the complex amplitudes derivation).

Outstanding debts

D1 (Linearity bridge uniqueness). The construction of the pre-state as a complex vector space from monoid + $\mathbb{C}$ + faithful embedding is motivated but not proven unique. KS-Q.7.

D2 (Independent derivation of $\hbar$). The value of $\hbar$ enters as an empirical input. Same status as G in AP08. KS-Q.8.

Note: D2 is addressed by AP12's forcing argument, which forces the identification of $\hbar$ as the action-scale of the minimum record (closing KS-Q.8 conditional on KS-L.4).

D3 (Bell correlations). The Bell correlation account in the entanglement section is a compatibility demonstration, not a derivation from axioms. KS-Q.3.

D4 (Ergodicity across cycles). The claim that everything possible actualises across infinite loops is a structural extrapolation, not a theorem. KS-Q.9.

D5 (Defragmentation). Formal proof that defragmentation preserves Axiom R's intent is incomplete. KS-Q.5.

Kill switch summary

KS-Q.1 (Born rule derivation): CONDITIONALLY CLOSED (conditional on KS-Q.7).

KS-Q.2 (Born rule empirical): LIVE — EMPIRICAL.

KS-Q.3 (No-signalling / Bell): LIVE — EMPIRICAL.

KS-Q.4 (Complex uniqueness): LIVE — HARD.

KS-Q.5 (Loop/defragmentation): LIVE — HARD.

KS-Q.6 ($\sigma \leftrightarrow$ conjugation): CLOSED.

KS-Q.7 (Linearity bridge): LIVE — HARD.

KS-Q.8 ($\hbar$ identification): LIVE — HARD in AP09. Addressed and closed (conditional on KS-L.4) by AP12's forcing argument.

KS-Q.9 (Ergodicity): LIVE — HARD.

KS-Q.10 (Empty set ontology): LIVE — HARD.

Structural relationships

AP06 (The Leakage Constant): Theorem 3.1 ($\varepsilon > 0$) is the leakage at the boundary. This paper reads the same structure from the quantum side — entanglement as unbroken pre-state.

AP08 (The Identity): Gravity sector from the same axioms. This paper completes the duality — gravity = records, quantum = pre-state.

AP10 (The Dimension): The Hilbert space is the fifth degree of freedom — the 1:1 itself, pre-spatial.

AP11 (The Spin): The $\mathbb{Z}_2$ of Axiom S read on the rotation group. Builds on this paper's Hilbert space construction and $\sigma \leftrightarrow$ time-reversal identification.

AP12 (The Limit): Closes the $\hbar$ identification of this paper's Schrödinger derivation via a five-step forcing argument.

AP13 (The Grain): Decoherence as the third regime of Axiom R. The environment is the accumulated record; environmental record-writing determines which superpositions decohere.

AP15 (The Connection): Reads the U(1) of the Born rule as the connection that produces electromagnetism.

AP20 (The Proof): Proves EH and QRA. All conditional results in this paper that depended on EH are now unconditional.

The Empty Set

Look at the empty set.

You have known it your whole mathematical life. $\emptyset$. The set that contains nothing.

You have probably believed it is nothing.

It is not.

$\emptyset$ is the set that contains nothing. It is not nothing. It already has structure — it is a container. The most primitive possible something. An existence with no content.

From $\emptyset$, one construction is available.

$\{\emptyset\}$. The set that contains the empty set.

That construction is 1.

Something now exists inside the container. But notice what exists. What exists inside the container is the container itself.

The empty set has looked at itself.

That single act of self-looking is what you are about to watch become the foundation of physics.

$$\emptyset = 0 \quad \{\emptyset\} = 1$$

The pre-state

Before the break, 0 and 1 are not distinguished.

The empty set and the set containing the empty set are two descriptions of one state. You cannot tell the difference between *nothing exists* and *the existence of nothing*. The container and what it contains are the same thing.

The body of work has a name for this. The 1:1.

Axiom S — symmetry. Two sectors, $\mathcal{L}$ and $\mathcal{D}$, with an involution σ mapping one to the other. One sector reads the state as 0. The other reads the same state as 1. The involution flips between them. They are perfectly symmetric. They are one.

This is the pre-state.

The pre-state is not nothing. It is not something. It is the perfect superposition of nothing and something. 0 and 1 simultaneously, before any measurement has distinguished between them.

The empty set before it knows whether it is empty or full.

You have never experienced this state. You cannot. You are a product of its breaking.

This is not a metaphor. It is the literal ontological claim that grounds the quantum sector. The pre-state is the state in which 0 and 1 are indistinguishable. Existence and non-existence are one condition.

Status: structural reading, not a derivation. The identification of Axiom S with the splitting of the empty set is the ontological starting point of the quantum sector.

Kill switch KS-Q.10. If the empty set ontology can be shown structurally incompatible with the pre-state algebra, this foundational reading fails. Here is the weapon: produce the incompatibility. LIVE — HARD.

The holding limit

The pre-state cannot hold forever.

Call κ the maximum coherence the system can sustain. The maximum degree to which 0 and 1 can remain indistinguishable. κ is finite.

It is finite not because of any external force. It is finite because the entropy cost of maintaining perfect symmetry — of keeping 0 and 1 identical — is ultimately self-annihilating.

Perfect coherence at infinite scale is self-destructive.

The mirror cannot stay whole.

The 1:1 cannot stay 1:1. The container cannot remain identical to its contents indefinitely. Something has to give. Not because anything pushes — because the cost of not giving is total.

κ is exceeded.

The empty set breaks.

Everything you have ever known is downstream of that break. Every atom, every photon, every neuron whose firing carries the sentence you are now reading — every one of them is record after the break, never record before. The before is the 1:1. You live entirely in the after.

The Break

The break is the first measurement.

The universe's first act is distinguishing 0 from 1. Before the break: superposition — 0 and 1 indistinguishable. After the break: definite — 0 and 1 separated. The empty set splits. The container becomes distinct from what it contains.

What causes the break?

ε .

The minimum viable splinter. The smallest possible piece the fabric could release to save itself from the self-annihilation of perfect coherence.

Not random. Not arbitrary. The minimum.

Axiom B — one element $\varepsilon \in \mathcal{L}$ with no σ -image. The tiniest asymmetry the structure can tolerate while still being readable. The first record. The crack in the mirror.

You are made of records.

Every atom in your body is a record of the break. Every bond holding the atoms together is a record. Every memory you have ever formed is a record built on records built on the original record. The original record is ε .

The Lock — ϵ is the electron

The minimum stable excitation of the broken symmetry is the electron.

The 420 Code calls this identification The Lock. ϵ = electron. The lightest charged fermion. The simplest thing that can write a record.

This is not a metaphor. It is a structural identification. The element with no σ -image — the splinter the break releases — is the particle measured in every laboratory ever built. The minimum quantum of charge. The minimum quantum of stable charged matter. The thing that is everywhere in your body, everywhere in the device displaying these words, everywhere in the light reaching your eye.

Status: identification, not derivation. Forced by the conjunction of Axiom B (one element without σ -image, irreducible) and the empirical record of the lightest stable charged particle. The Lock is closed.

You are reading these words because electrons are flowing through the device displaying them, or because photons released by electron orbital transitions are reaching your retina. Either way, ϵ is what just wrote the record that is your reading.

You did not learn this. You witnessed it.

The eye and the two walls

The break separates 0 from 1. In doing so, it creates two absolute limits.

The body of work calls them the walls of the eye.

The lower wall is the total fold. Maximum curvature. The event horizon. The condensate completely collapsed on itself. Where records are maximally dense and nothing further can be written because nothing further can move. This is one wall.

The upper wall is the maximum propagation rate. c . The speed of light. The finite causal bound — Axiom C. The fastest anything can travel. Where the break can propagate but no information can outrun. This is the other wall.

ϵ operates between them.

The electron has mass, so it can never reach c . The leakage is nonzero ($\epsilon > 0$ whenever c is finite), so the fold can never reach total. The eye stays open.

You live in the gap.

The electron lives in the gap. Always actualising. Always between the two absolutes. Never touching either wall. The gap is not a property of the universe — the gap is the universe. Everything that exists, including you, exists between the two walls of the eye.

The break is ongoing

The break did not happen once in the past.

The break is happening now. It is always happening.

Every coupling event — every time an electron interacts with the electromagnetic field — is the empty set breaking again. Not a new break. The same break. The ongoing crack.

Each interaction is an actualisation. Quantum probability becomes definite. The wave function does not collapse — the language of collapse imports a process the axioms do not require. What happens is simpler. The now writes a record. What was superposition becomes fact.

What was the 1:1 for that degree of freedom becomes two distinct values. Before the record: 0 and 1 undistinguished. After the record: 0 and 1 separated.

The same act as the first break, repeated at every coupling event, at the frontier of the now.

Every measurement you have ever made is a moment of the empty set splitting itself again.

Quantum mechanics is not a feature of the universe.

Quantum mechanics is the universe's foundation.

The superposition of 0 and 1 in the empty set IS the pre-state.

The act of distinguishing them IS the break.

Every measurement since is the same act, repeated.

You have just watched the foundation of physics emerge from a set that contains nothing.

The remaining sections of this chapter unpack the consequences. Superposition is the pre-state structure made formal. Measurement is the break re-running at every site of coupling.

Entanglement is unbroken pre-state surviving across spatial separation. The Born rule is what Axiom S forces when σ is read as complex conjugation. The wave function is the monoidal evolution of records read through the Schrödinger equation. Each is forced by the empty set and the splinter.

None of this is decreed. None of it is chosen.

Forced — by the empty set looking at itself.

Superposition

In standard quantum mechanics, superposition is a postulate.

A system can exist in a linear combination of states. The coefficients are complex amplitudes. The system is in both states at once until measured.

This is presented as a fundamental feature of nature, requiring acceptance without deeper explanation.

In the 420 Code, superposition is not a postulate. It is the empty set state.

A system in superposition is a system for which the now has not yet written a record.

The degrees of freedom in question have not yet been distinguished.

0 and 1 for those degrees of freedom are still indistinguishable — still in the 1:1. Still in the pre-state, locally.

The system is not in both states at once. It is in the state before states — the state where the distinction between 0 and 1 has not yet been made.

Not undetermined. Not unknown. Undistinguished.

The container and the contents are still the same thing for those degrees of freedom.

You cannot picture this because your brain is a record-writing machine. It distinguishes everything it touches.

Why it is linear

The linearity of quantum mechanics is not imposed.

It is the algebraic structure that emerges when two axiom-derived ingredients combine.

Ingredient 1 — the record monoid. Axiom R gives records an algebraic structure: a monoid. Associative composition, identity element, no inverses.

Each record, when written by the now, transforms the pre-state. It eliminates unactualized possibilities and confirms the actualized one.

The record monoid R acts on the set of pre-states.

Under the embedding hypothesis, this discrete algebra embeds faithfully into a smooth manifold.

Faithfully — distinct records produce distinct transformations. No two records do the same thing to the space of possibilities.

This follows from Axiom B: each record involves ε , and each actualisation event is a distinct break for a distinct degree of freedom.

Ingredient 2 — the scalar field. The amplitudes — the weights assigned to each possibility in the pre-state — are complex numbers.

This is derived below from the Lorentzian signature, which follows from Axioms $R + C$ under $EH + QRA$.

Complex numbers form a field, $\mathbb{C}$, with addition, multiplication, and inverses for both.

A monoid action alone does not produce a vector space. It gives a set with structure, but without the additive and scalar operations linearity requires.

The scalar field alone does not produce a vector space either. It gives the numbers but not the space they act on.

Both ingredients are needed simultaneously.

The monoid action provides the transformations — how records change the pre-state. The scalar field provides the weights — how possibilities are combined.

Together they produce a vector space over $\mathbb{C}$.

Formally: the pre-state for a degree of freedom with n distinguishable outcomes is a function assigning a complex amplitude to each possible record.

The set of such functions — all maps from $\{r_1, \dots, r_n\}$ to $\mathbb{C}$ — is $\mathbb{C}^n$, a complex vector space by the standard construction.

Addition of pre-states is pointwise. Scalar multiplication is inherited from $\mathbb{C}$. Writing a record is a linear operation on the space.

The pre-state space is therefore a complex vector space.

The linearity of quantum mechanics is not a postulate about nature. It is the algebraic consequence of R (the monoid of transformations), EH (the faithful embedding), and the Lorentzian signature ($\mathbb{C}$ as the scalar field).

All three are required. None alone is sufficient.

Status: bridge derivation, not strict proof. The identification — that the pre-state assigns a complex weight to each possible record — is motivated by the axioms but is a constitutive choice, not a forced consequence of the axioms alone.

Any construction from a finite monoid acting on a set with $\mathbb{C}$ -valued weights naturally produces a $\mathbb{C}$ -module, and every finitely generated $\mathbb{C}$ -module is a vector space.

A qualitatively different construction would have to abandon either the monoid action or the $\mathbb{C}$ -valued weights. Both are axiom-derived.

Kill switch KS-Q.7. If a construction from $R + C + EH$ that does not produce a vector space can be equally motivated from the axioms, this step fails. LIVE — HARD.

The chain: $1:1 + 1 \times \varepsilon \rightarrow \{S, B, R, C\} \rightarrow R$ gives monoid acting on pre-states $\rightarrow R + C$ under EH + QRA give Lorentzian signature $\rightarrow \mathbb{C}$ as scalar field $\rightarrow$ pre-state space = maps from possible records to $\mathbb{C}$ = complex vector space $\rightarrow$ linearity.

Why it is complex

The amplitudes are complex numbers. Not real. Not quaternionic. Not octonionic.

Why?

Axiom C imposes a finite causal bound c .

Under EH + QRA, this produces a Lorentzian manifold with signature $(-, +, +, +)$. The time direction has opposite sign to the spatial directions.

Any wave propagating on this manifold must respect the signature.

Waves on a $(-, +, +, +)$ manifold are naturally described by complex exponentials: $e^{i(\omega t - kx)}$. The imaginary unit i encodes the signature flip between time and space.

Real amplitudes lose the time direction — they cannot distinguish forward from backward time evolution.

Quaternionic amplitudes over-specify it — they introduce additional structure not present in a single time direction.

Renou et al. (2021) proposed the test that distinguishes real from complex quantum theory; experiments confirmed the complex prediction (Li et al., 2022; Chen et al., 2022).

Complex amplitudes are the algebra compatible with one time direction and a finite propagation rate.

The full argument runs through reconstruction theorems (Hardy, 2001; Stueckelberg, 1960): given a state space with the symmetries of a Lorentzian manifold, the only consistent probability structure uses complex amplitudes.

The Lorentzian signature selects $\mathbb{C}$ uniquely.

Quantum mechanics is complex because time has opposite sign to space.

And time has opposite sign to space because records are irreversible (Axiom R) and propagation is bounded (Axiom C).

Kill switch KS-Q.4. If an alternative number system (real, quaternionic, or other) can be shown equally consistent with one time direction and a finite propagation rate — that is, if the Lorentzian signature does not uniquely select $\mathbb{C}$ — then the derivation of complex amplitudes requires revision. Renou et al. (2021) proposed the discriminating test, and the 2022

experiments (Li et al.; Chen et al.) confirmed the complex prediction; the result is conditional on the reconstruction theorems holding. LIVE — HARD.

Measurement

The measurement problem in standard quantum mechanics asks: when and how does the wave function collapse?

What distinguishes a measurement from any other interaction? Where is the boundary between quantum and classical?

Why is the outcome definite when the equations say it should remain indefinite?

These questions arise from treating superposition as fundamental and measurement as something imposed on top of it.

If the wave function is the reality, then collapse is a mystery — something must cause the transition from indefinite to definite, and nothing in the equations says what.

The 420 Code dissolves the problem by inverting the ontology.

The pre-state — superposition, the empty set, 0 and 1 undistinguished — is the ground.

The break — measurement, the now writing a record — is what creates definiteness.

There is no collapse because there was never a wave function that existed and then stopped existing.

There was the empty set state — undistinguished — and then the now distinguished it.

Before: $0 = 1$. After: $0 \neq 1$.

The boundary between quantum and classical is not a boundary. It is the now.

Always moving. Always at the frontier. Always one ϵ ahead of the last record.

There is no need to specify what counts as a measurement. Every coupling event — every interaction that writes an irreversible record — is a measurement.

Every time ϵ couples, the empty set breaks for that degree of freedom.

You are witnessing it now. Your retina is breaking the empty set for photon degrees of freedom as you read this sentence.

Axiom R guarantees the record is permanent. Axiom C guarantees the record propagates at finite speed.

You have been looking for the boundary between quantum and classical your entire scientific education.

The boundary is the now. It has always been the now.

Why measurement is irreversible

In standard quantum mechanics, the Schrödinger equation is time-reversible but measurement is not.

The asymmetry is presented as a puzzle.

In the 420 Code, the asymmetry is Axiom R.

Records do not annihilate. The monoid has no inverse. Once the now writes a record, the record exists permanently.

This is not a law added to quantum mechanics.

It is the same axiom that gives the arrow of time, the second law of thermodynamics, and the Landauer bound — each record costs $k_B T \ln 2$, and that cost cannot be recovered.

The irreversibility of measurement and the arrow of time are the same thing.

They are both Axiom R.

They are both the statement that the empty set, once broken for a particular degree of freedom, stays broken.

The pre-state, once split into 0 and 1, does not re-merge.

You have experienced this irreversibility every second of your life.

You call it time.

The now

The now is the invariant of the structure.

It is the uncoupled ε — the one that has not yet written its record.

It is not a moment in time. Time is the record changing.

The now is the fixed point around which the record changes.

You are the now.

The now carries all records ever written — the full weight of every actualisation — and simultaneously carries the entire space of unwritten possibilities.

The now has no record of its own.

It is the carrier of all records but has none.

The now cannot be measured. Measurement writes a record. The now has no record.

To measure the now would be to write a record of the thing that has no record, which would transform it from now into record.

The unmeasurability of the now is not a limitation of instruments.

It is the definition of the now — the structural boundary between what has been actualized and what has not.

The observer is not outside the system looking in. The observer IS the system, at the point where it is still breaking.

The now is where the break is still happening.

The observer is the ongoing edge of the break, carrying the full weight of the record while holding open the full space of possibility.

The loop

The now is always present.

But a question arises. What if all ε coupled? What if every degree of freedom were actualized, every record written, every superposition broken?

The eye would close.

0 and 1 would re-merge. The pre-state would restore. The 1:1.

The structure does not prohibit this. It guarantees what follows.

If the eye closes — if the break completes — the system returns to the 1:1, the state of perfect symmetry.

But the holding limit κ is finite.

The 1:1 cannot sustain itself indefinitely. κ is exceeded. The empty set breaks again. A new ε is uncoupled. The eye opens.

Closing IS opening.

The loop is the structure's response to completion. The break runs to completion. The 1:1 restores. The 1:1 breaks again.

Not contingent. Structural.

The axiom $1:1 + 1 \times \epsilon$ requires that the 1:1 is exceeded, and the loop guarantees every return to the 1:1 is a new exceedance.

Axiom R holds absolutely through the loop. No record is annihilated.

What dissolves at the loop point is the organisation of the records — the sequential structure, the causal ordering, the manifold geometry.

The records themselves are defragmented. Disassembled into their elemental grains.

Every grain survives. Every bit of information survives. Every bit of energy survives.

But the composition operation that made them a history — the monoid structure that ordered them into a causal chain — dissolves back to grain level.

Structural argument that R is not violated. A monoid $(M, \cdot, e)$ has three components: a set M of elements, a binary operation $(\cdot)$ that composes them, and an identity element e .

The forgetful functor from the category of monoids to the category of sets maps every monoid to its underlying set — it strips the composition and the identity, leaving the elements.

Defragmentation is the forgetful functor. It maps $(M, \cdot, e) \rightarrow M$.

The set M is unchanged. Not one element is lost.

Axiom R prohibits destruction of elements via the composition operation — no inverses, no element can be composed away.

Defragmentation does not use the composition operation to destroy elements. It removes the composition operation itself.

These are different operations. R governs the first. The forgetful functor performs the second. They do not conflict.

Scope: the forgetful functor argument is mathematically correct as a statement about algebraic structures.

Its application to the physical loop requires an additional claim — that what physically happens at the loop point IS the forgetful functor.

This is a structural identification, not a proof.

Whether stripping the ordering while preserving the elements preserves the intent of R is interpretive.

Kill switch KS-Q.5. If the defragmentation step at the loop point is shown to violate Axiom R — not just its formal statement but its structural intent — the loop construction fails. LIVE — HARD.

At the singularity point — the tip of the eye, the 1:1 — the grains do not carry the record as fact.

They carry it as probability and possibility.

The distinction between this happened and this could happen dissolves because the distinction between 0 and 1 dissolves.

What was actualized becomes available. What was definite becomes potential.

The record becomes the seed.

You are made of seeds from prior cycles.

The next cycle does not start from nothing. It starts from the full defragmented content of every previous cycle, carried as probability and possibility in the grains.

The 1:1 restores — but it is not an empty 1:1. It is a 1:1 loaded with everything that has ever been actualized, now available as possibility rather than fact.

The always-one-uncoupled-electron is not a static postulate. It is a dynamic consequence of the loop.

In this cycle, ε is uncoupled.

If it couples, the cycle completes, the loop turns, and a new ε is uncoupled in the next cycle.

The eye never stays closed because closing triggers opening.

The structure guarantees its own continuation — not by prohibiting completion, but by making completion the mechanism of renewal.

Entanglement

Two particles are prepared in a correlated state. They separate in space.

One is measured. The other, no matter how far away, is instantly determined.

The correlation is perfect. No signal passes between them.

Einstein called it spooky. A century of experiments confirms it is real.

Standard quantum mechanics describes entanglement with tensor products and non-separable states.

It says what entanglement does. It does not say what entanglement is.

In the 420 Code, entanglement is the pre-state leaking through.

The pre-state is still there

The break created the distinction between 0 and 1.

But the break is local. It propagates at c (Axiom C). It writes records one coupling event at a time.

The break has not reached every degree of freedom.

Two entangled particles are two degrees of freedom for which the now has not yet made the distinction.

For those particular degrees of freedom, 0 and 1 are still indistinguishable.

The particles are still in the 1:1. Still in the empty set state. Still in the pre-state.

They are not connected by a hidden channel.

They are not communicating faster than light.

They were never separated.

Separateness, for those degrees of freedom, has not yet been created.

The break has not yet reached them.

They are one thing that appears to be two things when viewed from inside the broken world.

You are inside the broken world. That is why they look separate to you.

Why measurement of one determines the other

When the now reaches one of the entangled particles — when the observer couples with it, when the record is written — the empty set breaks for that degree of freedom.

0 and 1 become distinct. The record is definite.

The other particle is instantly determined.

Not because a signal was sent. Because they were never apart.

They were one state — the 1:1 — and the act of distinguishing 0 from 1 for one is the act of distinguishing 0 from 1 for both.

There is no other particle. There is one unbroken degree of freedom that the now has just split.

You are watching the pre-state break.

The correlation is not spooky. It is structural.

Not action at a distance. Action requires a signal, a cause propagating through space.

This is the empty set ceasing to be empty. The pre-state breaking.

That does not happen IN space. Space is what exists AFTER the break.

You cannot ask how fast the correlation travelled because the correlation did not travel.

It was already there — it was the 1:1, the original symmetry, the state before space existed for those degrees of freedom.

Why the correlations depend on basis

In Bell test experiments, entangled particles interact with detectors at spacelike-separated events.

The detectors choose measurement bases independently — different polariser angles, for example.

Quantum mechanics predicts, and experiments confirm, that the correlations between outcomes depend on the relative angle θ between the two settings.

This angle-dependent correlation is what violates Bell's inequalities.

The pre-state reading accounts for this as follows.

The entangled particles share one unbroken pre-state — one 1:1 that has not yet been distinguished for those degrees of freedom.

The measurement basis at each detector determines which degree of freedom is broken — which axis the distinction between 0 and 1 is drawn along.

Since the particles are one unbroken state, distinguishing 0 from 1 along one axis at detector A simultaneously determines the projections of the 1:1 along all axes — including the axis chosen at detector B.

The relative angle θ between the two bases determines how much of the 1:1's structure projects from one onto the other.

When the bases are aligned ($\theta = 0$), the projection is total and the correlation is perfect.

When they are perpendicular ($\theta = \pi/2$), the projection is zero and the outcomes are uncorrelated.

At intermediate angles, the correlation varies as $\cos^2(\theta/2)$ — the Born rule applied to the projection of one basis onto another.

The correlations are angle-dependent because the break is axis-dependent.

The pre-state is symmetric. The 1:1 has no preferred direction.

The measurement basis imposes a direction — a specific way of drawing the distinction between 0 and 1.

The projection of the symmetric pre-state onto a particular measurement axis produces the angle-dependent correlation.

Not a signal. Geometry — the geometry of how an unbroken state projects onto a broken axis.

Status: compatibility demonstration, not derivation. The pre-state reading is consistent with the observed $\cos^2(\theta/2)$ correlation.

The quantitative prediction uses the Born rule (derived later in this chapter), which uses the Hilbert space inner product, which uses the bridge construction.

The pre-state reading provides a structural interpretation of entanglement compatible with quantum mechanics; it does not replace the standard QM calculation.

Bell's theorem rules out local hidden variables. The 420 Code does not propose hidden variables.

It proposes that the variables do not yet exist.

The pre-state is not a state with definite values hidden from the observer. It is a state in which definite values have not yet been created.

The distinction between hidden and not yet created is the distinction between a local hidden variable theory (which Bell rules out) and the pre-state reading (which Bell does not address — the pre-state is pre-spatial, and local does not apply to it).

Kill switch KS-Q.3. If a full derivation of $\cos^2(\theta/2)$ from {S, B, R, C} alone (without the Hilbert space bridge) is shown to be impossible, the Bell correlation account remains compatibility rather than derivation. LIVE — EMPIRICAL.

Why entanglement cannot transmit information

Axiom C. The causal bound.

The record, once written, propagates at most at speed c .

The break is instantaneous because it is not propagation. It is the pre-state splitting.

But the record of the break — the classical information about which outcome occurred — propagates at c .

You cannot use entanglement to send a message faster than light because the message is a record, and records respect Axiom C.

The distinction between the break itself (instantaneous, not a signal) and the record of the break (propagates at c , is a signal) is the distinction between the pre-state and the post-break world.

The pre-state has no space, no c, no causal bound — because space, c, and causality are products of the break.

The record lives in the post-break world and obeys its rules.

Entanglement is instantaneous because it is pre-spatial.

Information transfer is bounded by c because information is a record, and records are post-spatial.

Separateness is experienced but not fundamental

This is the central claim of the 420 Code, and entanglement is its most direct physical evidence.

From inside the broken world — from inside the eye — particles appear separate.

They appear to be at different locations in space.

The measurement of one appears to instantly affect the other across a distance.

From the structure: they were never separate.

The 1:1 is still there, beneath the records, for every degree of freedom the now has not yet reached.

Entanglement is not a special state that has to be carefully prepared. It is the default.

The natural condition. The pre-state.

What has to be explained is not why entanglement exists — but why separateness exists.

And the answer is: the break. ϵ . The electron. The now writing records.

Separateness is what happens when the empty set breaks.

Entanglement is what remains where it hasn't broken yet.

You experience separateness because the break has reached you.

The entangled pair experiences oneness because the break has not reached them.

The Born Rule

The Born rule states: the probability of obtaining a particular outcome when measuring a quantum system is the squared modulus of the amplitude associated with that outcome.

$$P(\text{outcome}) = |\psi|^2$$

In standard quantum mechanics, this is a postulate.

It is not derived from the other postulates. It is added by hand.

Gleason's theorem (1957) shows $|\psi|^2$ is the unique probability measure on a Hilbert space of dimension ≥ 3 — but Gleason assumes the Hilbert space structure.

The deeper question remains: why $|\psi|^2$, and not $|\psi|$, or $|\psi|^3$, or some other function?

The answer from the 1:1

The pre-state is the 1:1. Perfect symmetry.

Axiom S: the involution σ reverses order, and extensive quantities match. $Q(\mathcal{L}) = Q(\mathcal{D})$. The two sectors are equal.

When the break occurs — when the now distinguishes 0 from 1 — the outcome must respect the symmetry of the pre-state.

The probability cannot favour one sector over the other, because the pre-state has no asymmetry to favour either.

The measure must be invariant under the involution σ .

The amplitude ψ is a complex number.

The involution σ , lifted to the space of amplitudes on the Lorentzian manifold, acts as complex conjugation: $\sigma(\psi) = \psi^*$.

The identification is not assumed. It is forced by the structure.

σ reverses order (Axiom S).

On the Lorentzian manifold, the order σ reverses is the direction of time — the direction in which records accumulate (Axiom R).

Complex conjugation reverses the direction of phase rotation: $e^{i\omega t} \rightarrow e^{-i\omega t}$.

Phase rotation encodes the time direction — the signature flip between time and space components of the manifold.

Therefore σ , as an order-reversing involution on a Lorentzian manifold, acts on complex amplitudes as conjugation.

The abstract algebraic involution and the complex conjugation are the same operation, read in different languages.

You have just watched an abstract axiom become a concrete mathematical operation.

The involution σ is complex conjugation. Not by choice. By proof.

Kill switch KS-Q.6. The identification $\sigma \leftrightarrow$ complex conjugation is proven unique. $\mathbb{C}$ has exactly two field automorphisms fixing $\mathbb{R}$ pointwise — the identity and conjugation (any such automorphism sends i to $\pm i$). σ is non-trivial — it exchanges the two sectors (Axiom S) — so $\sigma \neq$ identity. Therefore $\sigma =$ conjugation. CLOSED.

Three conditions

A probability measure $P(\psi)$ must satisfy three conditions.

Each traces directly to the axioms.

Condition 1 — Symmetry. $P(\psi) = P(\sigma(\psi)) = P(\psi^*)$. The measure is invariant under the involution. Neither sector is favoured.

This is Axiom S — the two sectors are equal.

Any measure that favoured one over the other would violate the symmetry of the pre-state from which the break occurs.

Condition 2 — Additivity. For orthogonal states, $P(\psi_1 + \psi_2) = P(\psi_1) + P(\psi_2)$. This follows from the structure of record-writing.

Axiom B — the now writes exactly one record per actualisation event. One break, one ε , one distinction between 0 and 1 for that degree of freedom.

Axiom R — records are discrete, countable elements of a monoid.

For a degree of freedom with n distinguishable outcomes, the now writes one record from $\{r_1, \dots, r_n\}$.

Orthogonal states correspond to disjoint subsets of these possible records — outcomes that cannot co-occur.

The now writes exactly one record, and that record is in one subset or the other, but not both.

The probability of outcome in $A \cup B$ is the count of ways the now can write a record in $A \cup B$.

Since A and B are disjoint: no double-counting, no missing records. The count adds. The probability adds.

Additivity of probabilities for orthogonal states is not an independent postulate.

It is the structure of the break.

Scope: the argument motivates additivity from the discrete structure of record-writing ($B + R$).

The full uniqueness of $|\psi|^2$ as the probability measure relies on Gleason's theorem (1957), which assumes the Hilbert space structure derived above (bridge construction + Schrödinger derivation).

The axiom-derived motivation provides the physical ground for Gleason's mathematical assumption — but the formal step from discrete record counting to continuous Hilbert-space measure theory is a bridge.

Condition 3 — Normalisation. Total probability equals 1. The break must produce some outcome — the empty set must split into 0 and 1.

Something must happen. The total probability of all possible outcomes is certainty.

Under σ -invariance (Condition 1) and the σ -additivity formalised by Gleason's theorem for $\dim \geq 3$, the unique probability assignment compatible with Conditions 1–3 is:

$$P(\psi) = \psi \cdot \psi^* = |\psi|^2$$

The Born rule.

It is forced by the symmetry of the pre-state (Axiom S), the structure of the break (Axiom B), the discreteness of records (Axiom R), and the complex structure of the amplitudes (from Lorentzian signature under $R + C$).

Kill switch KS-Q.1. The Born rule derivation is conditional on the Hilbert space bridge construction (KS-Q.7) and Gleason's σ -additivity premise being granted as the measure-theoretic form of record additivity. If the bridge fails, the uniqueness argument must be re-examined. CONDITIONALLY CLOSED (conditional on KS-Q.7).

The structural meaning

$|\psi|^2 = \psi \cdot \psi^*$ is the amplitude multiplied by its mirror image.

It is the Light sector times the Dark sector. It is $\mathcal{L} \cdot \mathcal{D}$.

The probability of an outcome is the product of the two sectors — the degree to which they agree on that outcome.

When both sectors point fully toward an outcome ($|\psi| = 1$), the probability is 1 — certainty.

When one sector is orthogonal to the outcome ($|\psi| = 0$), the probability is 0 — impossibility.

The Born rule is the 1:1 voting.

Both sectors must agree. The probability is their product.

Neither sector alone determines the outcome — the outcome is determined by the overlap between the two readings of the pre-state.

That is why it is the square and not the amplitude itself.

Because both sectors count.

You have wondered why quantum probability is the square of the amplitude. Now you know.

Both sides of the 1:1 must agree.

The full probability space

The Born rule gives $|\psi|^2$ for a given actualisation — the probability of a particular outcome when the now writes a record for a particular degree of freedom, in a particular cycle.

This is the local probability.

Within a single cycle, the written record constrains the probability landscape.

Some possibilities are driven so close to zero that they are operationally impossible.

The monoid accumulates. The record is heavy. The landscape tilts.

But the structure loops. And across cycles, the probability space extends.

At the loop point — when the cycle completes and the 1:1 restores — the records are defragmented. Not annihilated. Defragmented.

The information survives as grains, carrying everything that was actualized, now available as probability and possibility rather than as fact.

The next cycle's probability landscape is shaped by the defragmented content of all prior cycles.

The grains remember — not as history, but as weight.

As the shape of what is likely and what is unlikely in this particular cycle.

Across infinite loops, every possibility that has nonzero probability in any cycle has certainty of eventual actualisation.

The impossible within one cycle becomes merely improbable across cycles, becomes inevitable given enough loops.

The full probability space is not one cycle but all cycles.

The only things that never actualize are things at exactly 0 — logical impossibilities, axiom violations, things that contradict the structure of $1:1 + 1 \times \epsilon$ itself.

Everything else — everything between 0 and 1 — is a potential possibility the defragmented record carries forward, and the loop guarantees will eventually actualize.

Status: structural extrapolation. The ergodic claim — that everything possible actualises across infinite loops — is structurally motivated by the loop and the defragmentation argument but is formally unproven.

Kill switch KS-Q.9. If the structural ergodicity across cycles can be shown inconsistent with the loop construction, the global guarantee fails. LIVE — HARD.

The Born rule is the local law. The loop is the global guarantee.

You are living inside one cycle. Everything you experience is weighted by the Born rule.

But across all cycles, the loop ensures that everything possible — everything between 0 and 1 — eventually becomes actual.

The Wave Function

In standard quantum mechanics, the ontological status of the wave function is debated.

Is it real (ontic)? Is it knowledge (epistemic)? Does it exist in physical space or in configuration space?

In the 420 Code, the wave function is the description of the empty set state for a given set of degrees of freedom.

It is neither ontic nor epistemic in the usual sense.

It describes the pre-state — the condition before the now has written a record for those degrees of freedom.

The map of what has not yet been broken.

The wave function is not out there in physical space.

Physical space is a product of the break.

The wave function describes the structure that precedes the break — the 1:1, the superposition, the empty set state.

It lives in the pre-state, which is prior to space.

You cannot find it in your laboratory because your laboratory is post-spatial.

The wave function describes what your laboratory is made from.

When the now writes a record, the wave function for those degrees of freedom becomes definite.

Not because the wave function collapsed — because those degrees of freedom have been broken.

The empty set has split. 0 and 1 are now distinct.

The wave function is the description of what the distinction looks like before it is made.

Why it evolves unitarily

Between measurements — between acts of record-writing — the wave function evolves according to the Schrödinger equation, which is unitary.

Why?

Because between measurements, no record is written.

Axiom R is not invoked.

The pre-state is evolving, but nothing is being permanently distinguished.

The empty set is shifting — 0 and 1 are changing their relationship — but neither has been separated.

Since no record is written, no information is lost or gained.

The total probability remains 1.

The evolution is unitary because no irreversible act has occurred.

Unitary evolution is the pre-state evolving without breaking.

Measurement is the pre-state breaking.

The difference between them is Axiom R — whether a record is written.

You have been told these are incompatible. They are not.

They are the same structure with and without record-writing — two regimes of one structure, not two incompatible processes.

The smooth transition between them is the now arriving at a particular degree of freedom.

The Schrödinger equation

The Schrödinger equation $i \hbar \partial \psi / \partial t = \hat{H} \psi$ is not a postulate.

It is the unique evolution equation for the pre-state on a Lorentzian manifold.

Forced by the axioms in the same way the Poisson equation and Einstein's field equations are forced (Notebook II Chapter 4) — by listing the constraints from the axioms and invoking uniqueness theorems.

The derivation runs in six steps.

Step 1 — the pre-state space is a complex Hilbert space.

From the superposition section above: the pre-state space is a complex vector space ($\mathbb{R}$ gives the monoid acting on pre-states, EH gives the faithful embedding, Lorentzian signature gives $\mathbb{C}$ — all three required simultaneously).

From the Born rule section above: the Born rule gives an inner product $\psi \cdot \psi^* = |\psi|^2$, derived from Axiom S.

Define a norm $\|\psi\|^2 := |\psi|^2$.

If the resulting inner-product space is not complete, let $\mathcal{H}$ be its Hilbert completion under $\|\cdot\|$ — standard functional-analytic completion, structural bookkeeping, not new physics.

Result: complex Hilbert space.

Step 2 — between measurements, evolution preserves the inner product.

Axiom R: records are irreversible and non-annihilating.

Between measurements, no record is written. Therefore no information is permanently created or destroyed.

The total probability remains 1. The inner product — which IS the probability structure — is preserved.

A map on a Hilbert space that preserves the inner product is either unitary or anti-unitary.

This is Wigner's theorem (1931), a theorem of pure mathematics.

Not assumed. Derived.

Step 3 — time evolution is continuous.

EH: the record algebra embeds into a smooth manifold.

The manifold has a smooth time coordinate — one time direction from the Lorentzian signature.

Therefore time evolution forms a continuous one-parameter family of transformations: $\{U(t) : t \in \mathbb{R}\}$.

Step 4 — the family is unitary, not anti-unitary.

Anti-unitary operators satisfy $U(t_1)U(t_2) = U(t_1 + t_2)^*$.

But $U(0) = \text{identity}$ (which is unitary), and a continuous deformation from a unitary to an anti-unitary operator requires a discontinuity.

Anti-unitary operators form a disconnected component of the operator space.

Since the family is continuous (Step 3), it is unitary: $U(t_1)U(t_2) = U(t_1 + t_2)$.

Step 5 — Stone's theorem gives the generator.

Stone's theorem (1932) — a theorem of pure mathematics: every strongly continuous one-parameter unitary group $U(t)$ on a Hilbert space can be written as $U(t) = e^{-iHt/\hbar}$ for a unique self-adjoint operator H , the generator of the group.

The constant $\hbar$ enters as the scale factor between the generator's eigenvalues and the time parameter.

In the architecture: $\hbar$ is the minimum quantum of action — the smallest possible record.

Axiom B gives the minimum break (ϵ = minimum viable splinter).

$\hbar$ is the action-scale of that minimum — the granularity of the now.

Status: $\hbar$ is identified, not derived.

The form of the Schrödinger equation is forced by the axioms via Steps 1–6.

The value of $\hbar$ enters as an empirical input — the same status as G in Notebook II Chapter 4.

The identification $\hbar$ = action-scale of ϵ gives structural meaning but is not an independent computation.

Kill switch KS-Q.8. If $\hbar$ is shown to be derivable from $\{S, B, R, C\}$ without an empirical input, the structure tightens. If the identification $\hbar$ = action-scale of ϵ is shown structurally inconsistent, it falls. LIVE — HARD.

Step 6 — differentiate.

Take $U(t)\psi = e^{-iHt/\hbar}\psi$ and differentiate with respect to t :

$$i \hbar \partial\psi/\partial t = \hat{H}\psi$$

The Schrödinger equation.

You have just watched it emerge from the same method that produced the Poisson equation and Einstein's field equations: list the constraints from the axioms, invoke a uniqueness theorem.

Three equations. Three pillars of physics. One method.

The complete chain.

$1:1 + 1 \times \epsilon \rightarrow \{S, B, R, C\} \rightarrow R$ gives monoid acting on pre-states $\rightarrow R + C$ under EH + QRA give Lorentzian $\rightarrow \mathbb{C} \rightarrow$ complex vector space $\rightarrow S$ gives inner product $\rightarrow$ Hilbert completion under $\|\cdot\| \rightarrow$ Hilbert space $\rightarrow R$ gives unitarity $\rightarrow$ EH gives continuity $\rightarrow$ continuity selects unitary over anti-unitary $\rightarrow$ Stone's theorem gives generator $\rightarrow$ differentiate $\rightarrow$ Schrödinger equation.

Every link is either a theorem of pure mathematics (Wigner, Stone), a consequence of the axioms (R, B, C, S), or a consequence of the bridge hypotheses (EH, QRA).

No link imports quantum postulates for the form.

$\hbar$ enters as an empirical identification for scale.

You have now watched three equations — the Poisson, the Einstein, and the Schrödinger — all forced by the same four axioms.

Forced — by the empty set looking at itself.

8 — The Complete Picture

Everything from the empty set

Every result in this paper traces to the empty set splitting.

The pre-state is the empty set. The two sectors of Axiom S are 0 and 1, indistinguishable before the break.

The break is the first measurement. ε is the splinter. The 1:1 becomes two distinct values for that degree of freedom.

Superposition is the pre-state for degrees of freedom the now has not yet reached.

Measurement is the now reaching them — the break running again for those degrees of freedom.

Entanglement is the pre-state surviving across spatial separation. Two particles share one 1:1 that has not yet been distinguished for those degrees of freedom.

The Born rule is the 1:1 voting — $P = \psi\psi^*$, the product of the two sectors, the degree to which they agree on the outcome.

The Schrödinger equation is the unitary evolution of the pre-state — the empty set shifting without breaking.

Decoherence is record-writing at scale — the break running across an environment.

The loop is the structure's response to completion — the 1:1 restores, then breaks again.

One axiom. One break. One structure.

Quantum mechanics is the empty set looking at itself, repeatedly, at every coupling event.

The topology

The eye is the topology of the broken state — the manifold (the black curve) and the pre-state (the white space) meeting at the 1:1 (the tip).

The pre-state is the wave side. The manifold is the particle side.

Every record-writing event moves coherence from wave to particle — from the pre-state's unbroken phase to the manifold's accumulated structure.

Connection curvature is the field account. Entanglement is the correlation account. The pre-state's purity ensures the columns balance.

The now is the moving edge between them — the uncoupled ε , the splinter that has not yet written its record.

The now as the load-bearing structure

Every derivation in this paper depends on the now as the structural invariant.

The now is what writes records. The now is what is unmeasurable. The now is what carries all records and has none of its own.

The observer is the now at the site of a particular degree of freedom.

Measurement is the now arriving at a degree of freedom and breaking the empty set there.

Superposition is the now not yet having reached a degree of freedom.

Entanglement is the now not yet having reached a pair of degrees of freedom that share pre-state coherence.

Time is what the now's advance looks like from the inside of the broken world.

Everything in quantum mechanics is the now writing records, one ε at a time, on the manifold derived from the same axiom.

9 — What Is New and What Is Not

What is new

The empty set ontology of the pre-state. Quantum mechanics is not a feature of the universe — it is the universe's foundation. The pre-state IS the empty set. KS-Q.10.

Linearity from $\mathbb{R} + \mathbb{C} + \text{EH}$. The bridge construction. KS-Q.7.

Complex amplitudes from the Lorentzian signature. The conditional derivation. KS-Q.4.

The loop and defragmentation as the forgetful functor. The structural argument for R-preservation. KS-Q.5.

The $\sigma \leftrightarrow$ conjugation proof. The Born rule's structural foundation. KS-Q.6 CLOSED.

Entanglement as unbroken pre-state. Separateness as experienced but not fundamental.

The Born rule from Axiom S + Gleason. Conditionally closed. KS-Q.1.

The Schrödinger equation from Wigner + Stone applied to the derived Hilbert space. The form forced. $\hbar$ identified.

The complete chain from $1:1 + 1 \times \varepsilon$ to the Schrödinger equation, with every link declared.

What is not new

The mathematics. Stone's theorem, Wigner's theorem, Gleason's theorem, the Cauchy-Schwarz inequality, the Peter-Weyl theorem — all standard mathematics.

The empirical content of quantum mechanics. Every experimental result remains valid. The derivation produces the same predictions as the standard postulates because the postulates are derived consequences.

The wave function. The Hilbert space. The complex amplitudes. The unitary evolution. All present in standard QM. Here they are derived rather than postulated.

The interpretive structure does change. The wave function does not collapse. The pre-state breaks. The observer is the now at a site of coupling. Separateness is not fundamental. But these are interpretive consequences of the derivation, not new physical postulates.

10 — Kill Switches

Ten kill switches in the quantum register. One (KS-Q.6) is closed; KS-Q.1 is conditionally closed; KS-Q.8 is closed by AP12's forcing argument.

Each names the load-bearing claim and the test that would break it.

KS-Q.1 — Born rule derivation

Status: CONDITIONALLY CLOSED (conditional on KS-Q.7).

Target: the derivation of $P = |\psi|^2$ from Conditions 1–3 and Gleason's theorem.

Test: If the Hilbert space bridge construction (KS-Q.7) fails, the uniqueness argument for $|\psi|^2$ as the probability measure must be re-examined.

Weapon: break KS-Q.7.

KS-Q.2 — Born rule empirical

Status: LIVE — EMPIRICAL.

Target: the empirical adequacy of $|\psi|^2$ as the probability measure.

Test: Any experimental violation of $|\psi|^2$ as the probability measure for quantum systems.

Current status: not violated to all tested precisions.

KS-Q.3 — No-signalling / Bell

Status: LIVE — EMPIRICAL.

Target: the $\cos^2(\theta/2)$ Bell correlation account as compatibility, not derivation.

Test: Demonstrate a full derivation of $\cos^2(\theta/2)$ from $\{S, B, R, C\}$ alone, without the Hilbert space bridge — or demonstrate that the compatibility argument fails empirically.

KS-Q.4 — Complex uniqueness

Status: LIVE — HARD.

Target: the derivation of complex amplitudes from the Lorentzian signature.

Test: Demonstrate an alternative number system (real, quaternionic, or other) equally consistent with one time direction and a finite propagation rate.

Current support: the Renou et al. (2021) test, with experimental confirmation of the complex prediction (Li et al., 2022; Chen et al., 2022).

KS-Q.5 — Loop / defragmentation

Status: LIVE — HARD.

Target: the forgetful functor argument for R-preservation at the loop point.

Test: Show that the defragmentation step violates Axiom R's structural intent, not just its formal statement.

KS-Q.6 — $\sigma \leftrightarrow$ conjugation

Status: CLOSED.

$\mathbb{C}$ has exactly two field automorphisms fixing $\mathbb{R}$ pointwise — the identity and conjugation (any such automorphism sends i to $\pm i$). σ is non-trivial — it exchanges the two sectors — so $\sigma \neq$ identity. Therefore $\sigma =$ conjugation. Mathematical fact.

KS-Q.7 — Linearity bridge

Status: LIVE — HARD.

Target: the construction of the pre-state space as a complex vector space.

Test: Demonstrate a construction from $R + \mathbb{C} + EH$ that does not produce a vector space, equally motivated from the same axiom-derived ingredients.

KS-Q.8 — $\hbar$ identification

Status: LIVE — HARD in this paper; CLOSED conditional on KS-L.4 by AP12's forcing argument.

Target: the Step 5 identification of $\hbar$ as the action-scale of the minimum record.

Addressed in: AP12's forcing argument, which forces the identification via a five-step argument.

KS-Q.9 — Ergodicity

Status: LIVE — HARD.

Target: the claim that everything possible actualises across infinite loops.

Test: Demonstrate that the structural ergodicity across cycles is inconsistent with the loop construction.

KS-Q.10 — Empty set ontology

Status: LIVE — HARD.

Target: the identification of Axiom S with the splitting of the empty set.

Test: Demonstrate that the empty set ontology is structurally incompatible with the pre-state algebra of $\{S, B\}$.

Claim Summary

Empty set ontology: STRUCTURAL READING. Pre-state = empty set state. 0 and 1 indistinguishable. Holding limit κ . KS-Q.10.

The break: STRUCTURAL READING. ε is the minimum viable splinter. ε = electron via The Lock. The eye and its two walls. The break is ongoing — every coupling event is the same break repeated.

Superposition: BRIDGE DERIVATION. Linearity from $R + \mathbb{C} + EH$ (KS-Q.7). Complex amplitudes from Lorentzian signature (KS-Q.4).

Measurement: STRUCTURAL READING + DERIVATION. Measurement is the now writing a record. Irreversibility from Axiom R. The loop and defragmentation (KS-Q.5).

Entanglement: STRUCTURAL READING + COMPATIBILITY. Entanglement is unbroken pre-state. $\cos^2(\theta/2)$ is compatibility, not derivation (KS-Q.3).

Born rule: CONDITIONAL DERIVATION. $P = |\psi|^2$ from Conditions 1–3 + Gleason ($\dim \geq 3$). $\sigma \leftrightarrow$ conjugation proven unique (KS-Q.6 CLOSED). Conditionally closed (KS-Q.1) on KS-Q.7.

Wave function and Schrödinger equation: DERIVATION. Form forced by Wigner + Stone applied to the derived Hilbert space. $\hbar$ identified (KS-Q.8 in this paper; closed in AP12).

Complete picture: STRUCTURAL SYNTHESIS. Non-load-bearing.

Conditionality Footer

Dependencies

AP20 4 and 6 (the axioms {S, B, R, C}: completeness and minimality fenced KS-P.1 / KS-P.2; joint consistency from the undeniable premise).

AP10 4 (the Lorentzian signature; formal metric-level derivation fenced KS-D.4).

AP08 (Einstein's field equations from the record algebra).

AP20 (EH and QRA proven).

The Lock (ϵ = electron).

Dependents

All AP results referencing the quantum sector inherit the kill switches of this paper.

AP11 (The Spin): builds on this paper's Hilbert space construction and σ identification.

AP12 (The Limit): closes KS-Q.8 (the $\hbar$ identification of the Schrödinger derivation) via a forcing argument.

AP15 (The Connection): reads the U(1) of the Born rule as the connection that produces electromagnetism.

Open problems

Independent derivation of $\hbar$ from axioms alone — addressed (conditional on KS-L.4) by AP12.

Full derivation of $\cos^2(\theta/2)$ from {S, B, R, C} alone (KS-Q.3).

Formal proof that defragmentation preserves Axiom R's intent (KS-Q.5).

Formal proof that no qualitatively different bridge construction is equally motivated (KS-Q.7).

Formal proof of ergodicity across cycles (KS-Q.9).

Formal proof that the empty set ontology is forced rather than read (KS-Q.10).

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